

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459505.6454 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.049
Transit depth (Rp/Rs)^2	2.04 +/- 1.41 [%]
Orbital Inclination (inc)	88.47 +/- 2.89
Airmass coefficient 1 (a1)	3.89 +/- 0.74
Airmass coefficient 2 (a2)	-1.09 +/- 0.99
Scatter in the residuals of the lightcurve fit is	18.53 %
Best Comparison Star	#1 - [437, 139]
Optimal Aperture	1.8
Optimal Annulus	11.81
Transit Duration (day)	0.094 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.049 vs 0.1618 (deviation 0.38σ)
Duration fit vs published	0.094 d vs 0.0721 d (deviation 0.95σ)
Mid-time O–C	-18.7 min
Depth SNR	2.9
Residual scatter	18.53 %

Light curve

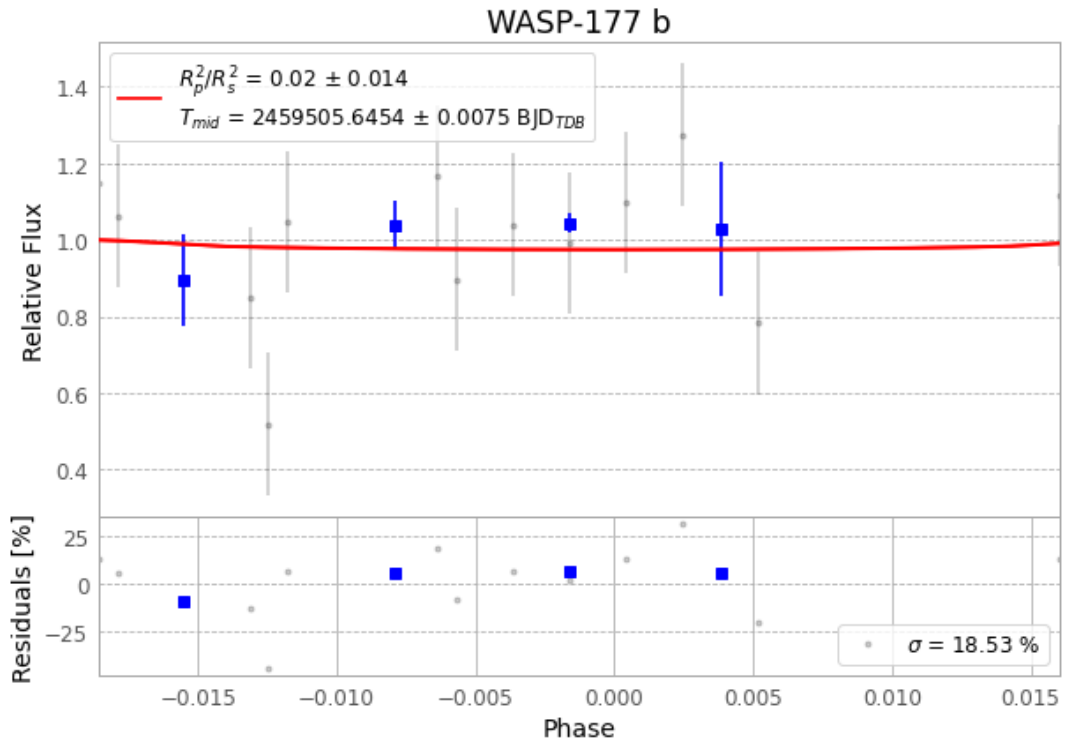


Figure 1 — FinalLightCurve_WASP-177 b_2021-10-17.png (SHA-256 b4b54109e005126f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 77], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2778c59db36c95...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

65 FITS frames (43 MB), WASP-177211018020012.FITS → WASP-177211018051212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-177-211018.log (8045dc19b7cf...), FinalLightCurve_WASP-177 b_2021-10-17.png (b4b54109e005...), FinalLightCurve_WASP-177 b_2021-10-17.csv (3a6c0c9c72a5...)

Compute resources

Wall time 174.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 01:08Z.